NAGPUR PULSE

AI/ML Engineering — 24-Hour Hackathon Execution Report

Revision 2 — Integrated Accuracy & Junction-Based Training Strategy

ROLE	TEAM SIZE	DURATION	PRIMARY AI GOAL
AI/ML Engineer / AI Integration Owner	5 Members	24 Hours	Reliable junction-level LOW/MEDIUM/HIGH prediction

REVISION APPLIED

The final junction feature set, multiclass target, leakage-safe splitting, TimeSeriesSplit / GroupKFold, 5-fold validation, Logistic Regression baseline, Random Forest comparison, XGBoost primary model, hyperparameter tuning, probability calibration, confidence-based UNCERTAIN output, and HIGH-risk precision ≥90% safety target are integrated into the AI/ML execution strategy.

1. Updated Executive Summary

NAGPUR PULSE is a smart traffic and public-safety decision-support system for junction-level risk intelligence. The AI module combines junction geometry, live traffic behavior, safety indicators, weather/road conditions, and rolling historical accident features.

```
Junction + Traffic + Safety + Weather/Road + Historical Risk
                         ↓
                  Data Cleaning
                         ↓
               Feature Engineering
                         ↓
             Leakage-Safe Split
          Time-Based / Group-Based
                         ↓
                  Preprocessing
                         ↓
       Logistic Regression / Random Forest / XGBoost
                         ↓
                 5-Fold Validation
                         ↓
                Hyperparameter Tuning
                         ↓
                   Best Model
                         ↓
               Probability Calibration
                         ↓
                 Confidence Threshold
                         ↓
             LOW / MEDIUM / HIGH
                         ↓
             Low Confidence → UNCERTAIN
                         ↓
              Explainability → FastAPI
                         ↓
              Backend → Dashboard
```

SAFETY-FOCUSED ACCURACY TARGET

HIGH-RISK PRECISION ≥ 90% is the primary safety metric. This is a validation target, not a guarantee of real-world safety. The threshold must be selected using training/validation data and frozen before the final held-out test.

2. Final Junction-Based Feature Set

GROUP	CANONICAL FEATURES
Junction	junction_id; junction_type; number_of_arms; number_of_lanes; road_width; intersection_angle; turning_lanes; median_present; signalized; signal_cycle_time; pedestrian_crossing; speed_limit
Traffic	traffic_volume; average_speed; speed_variance; traffic_density; queue_length; heavy_vehicle_percentage; two_wheeler_percentage; turning_vehicle_percentage; peak_hour
Safety	overspeeding_percentage; sudden_braking_count; lane_change_frequency; wrong_side_driving_count; pedestrian_volume; near_miss_count
Weather & Road	rainfall; visibility; temperature; weather_condition; wet_road; road_surface_condition; lighting_condition
Historical Risk	accidents_7d; accidents_30d; accidents_90d; accidents_1y; fatal_accidents_1y; injury_accidents_1y; historical_accident_rate

These names and meanings should be frozen in feature_config.json and used identically during training and FastAPI inference.

Feature quality rules

• Rolling historical features must only use information available before the prediction timestamp.

• Traffic and safety features must be timestamp-aligned with the prediction event.

• Categorical encoders and preprocessing must be fitted only on training data.

• junction_id is the grouping identifier; do not use naive leakage-prone encoding when evaluating unseen junctions.

3. Target & Prediction Output

Training target:

```
LOW
MEDIUM
HIGH
```

Prediction output:

```
LOW
MEDIUM
HIGH
UNCERTAIN
```

UNCERTAIN IS NOT A TRAINING CLASS

UNCERTAIN is an inference-time state produced when calibrated confidence is below the configured threshold or input quality is insufficient. The model can remain a three-class classifier.

The API should return riskLevel, confidence, class probabilities, topFactors, modelVersion, and timestamp.

4. Target Generation & Label Governance

If authoritative LOW/MEDIUM/HIGH labels are unavailable, define a deterministic prototype labelling rule before training. A normalized combination of accident frequency and severity indicators can be used, but the exact rule and thresholds must be versioned.

NO LABEL LEAKAGE

Do not calculate target thresholds, normalization statistics, or historical features using the complete dataset before splitting. Any information that would be unavailable at prediction time must stay out of the training features.

5. Model Strategy

MODEL	ROLE	SELECTION PURPOSE
Logistic Regression	Baseline	Simple benchmark and sanity check
Random Forest	Comparison	Nonlinear tree-based alternative
XGBoost	Primary	Preferred final candidate for structured tabular data

XGBoost is the primary model, but the final choice must be evidence-based. Select the best model using the predefined validation protocol and safety metrics rather than assuming XGBoost will always win.

6. Dataset Split & Cross-Validation

Known junctions: use time-based validation so past observations train the model and future observations test it.

```
Past Data
    ↓
TRAIN
    ↓
Future Data
    ↓
TEST

Example:
Jan–Sep → Train
Oct–Dec → Test
```

Unseen junctions: use GroupKFold with group = junction_id. This prevents the same junction from appearing in both training and validation.

USE CASE	VALIDATION	PURPOSE
Known junction / future prediction	TimeSeriesSplit / time-aware folds	Realistic temporal generalization
Unseen junction	GroupKFold(group=junction_id)	Generalization to new junctions
Final report	Untouched future test set	Unbiased final estimate

Use 5-fold cross-validation where data volume permits. All preprocessing and tuning remain inside the training/validation process.

7. Evaluation Metrics & Safety Gate

METRIC	USE
Accuracy	Overall correctness
Precision	Reliability of predicted class
Recall	Ability to capture actual class
F1-score	Precision/recall balance
HIGH-risk Precision	Primary safety metric
HIGH-risk Recall	Supporting safety metric
Confusion Matrix	Class-specific error analysis
ROC-AUC	Supporting discrimination metric where appropriate

PRIMARY SAFETY METRIC

HIGH-RISK PRECISION ≥ 90%. Report HIGH precision and HIGH recall separately. Also report macro/weighted F1 so strong performance on one class cannot hide poor performance on another.

Do not present a generic '95% accuracy' figure. Every metric must state dataset, split, target definition, validation method, and test protocol.

8. Hyperparameter Tuning

MODEL	PARAMETERS TO CONSIDER
Logistic Regression	C; penalty/solver; class_weight
Random Forest	n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features; class_weight
XGBoost	n_estimators; max_depth; learning_rate; subsample; colsample_bytree; min_child_weight; reg_alpha; reg_lambda

For a 24-hour hackathon, use a controlled randomized/grid search. Optimize against validation performance with HIGH-risk precision explicitly monitored. Never tune on the final test set.

9. Class Imbalance & HIGH-Risk Threshold

Check LOW/MEDIUM/HIGH class counts before training. If HIGH is underrepresented, use appropriate class weighting or other leakage-safe strategies.

The final HIGH decision threshold should be selected on validation data to achieve the safety objective while retaining acceptable HIGH recall. Freeze the threshold before final test evaluation.

IMPORTANT

A ≥90% HIGH-risk precision target is a decision-quality target. It is not evidence that the model is 90% safe in real-world operation. Prospective validation on representative operational data is required before production use.

10. Probability Calibration & Confidence

```
Model probabilities
        ↓
Probability Calibration
        ↓
Calibrated P(LOW), P(MEDIUM), P(HIGH)
        ↓
Confidence = maximum calibrated probability
        ↓
Confidence Threshold
        ↓
Enough confidence → LOW / MEDIUM / HIGH
Low confidence → UNCERTAIN
```

Calibration must be fitted using validation data and must not use the final held-out test set. Store the calibration artifact with the model version.

11. Final ML Pipeline

```
Junction Data
      ↓
Data Cleaning
      ↓
Feature Engineering
      ↓
Time-Based / Group Split
      ↓
Preprocessing
      ↓
Logistic Regression
Random Forest
XGBoost
      ↓
5-Fold Validation
      ↓
Hyperparameter Tuning
      ↓
```

```
Best Model
      ↓
Probability Calibration
      ↓
Confidence Threshold
      ↓
LOW / MEDIUM / HIGH
      ↓
Low Confidence → UNCERTAIN
```

12. Updated Risk API Contract

```
{
  "locationId": "ITWARI_001",
  "riskLevel": "HIGH",
  "confidence": 0.93,
  "probabilities": {
    "LOW": 0.01,
    "MEDIUM": 0.06,
    "HIGH": 0.93
  },
  "topFactors": [
    "near_miss_count",
    "overspeeding_percentage",
    "accidents_30d"
  ],
  "modelVersion": "risk-model-v2",
  "timestamp": "2026-08-17T..."
}
```

The earlier normalized 0–1 risk score can remain as a dashboard severity representation if required, but it must not be described as an accident probability unless formally calibrated as such.

13. Explainability

SHAP remains the preferred explanation method for XGBoost. Feature importance or model-appropriate contribution methods are the fallback for comparison models.

• Show the top 3–5 actual contributing features.

• Use the final canonical feature names.

• Do not hard-code generic explanations that are not supported by the current prediction.

• Show confidence alongside the explanation.

```
ITWARI JUNCTION
Risk Level: HIGH
Confidence: 0.93

Top contributors:
+ near_miss_count
+ overspeeding_percentage
+ accidents_30d
```

14. Updated 24-Hour Execution Plan

TIME	ML TASK	DELIVERABLE
H+0–H+1	Data + junction schema lock	Historical data; junction master; feature availability
H+1–H+2	Cleaning + temporal features	Validated feature table
H+2–H+3	Target + class distribution	LOW/MEDIUM/HIGH labels
H+3–H+4	Baseline models	Logistic + initial RF/XGBoost
H+4–H+5	5-fold/time-aware validation	Comparable metric table
H+5–H+6	Hyperparameter tuning	Best candidate

TIME	ML TASK	DELIVERABLE
H+6–H+7	HIGH threshold analysis	Precision/recall trade-off
H+7–H+8	Probability calibration	Calibrated probabilities
H+8–H+9	Model V2 freeze	Model + preprocessing + configs
H+9–H+10	Explainability	Top factors
H+10–H+11	FastAPI	Health + prediction endpoints
H+11–H+13	API tests	Single/batch/invalid/confidence tests
H+13–H+15	Backend integration	Node.js → FastAPI
H+15–H+16	Deployment	Reachable service
H+16–H+17	Full integration	Dashboard receives class/confidence
H+17–H+19	Dynamic demo	Traffic/incident → prediction change
H+19–H+21	Robustness	Fallbacks + low confidence
H+21–H+22	Final metrics/docs	Evaluation table + limitations
H+22–H+23	Rehearsal	Three successful runs
H+23–H+24	Freeze	No new features; backup

15. Model Artifacts & Reproducibility

```
models/
■■■ risk_model.joblib
■■■ preprocessor.joblib
■■■ calibration.joblib
■■■ feature_config.json
■■■ label_config.json
■■■ threshold_config.json
■■■ metrics.json
```

metrics.json should record the split dates/groups, class distribution, accuracy, precision, recall, F1, HIGH precision/recall, confusion matrix, ROC-AUC where applicable, calibration information, and final confidence/decision threshold.

16. FastAPI & Backend Integration

ENDPOINT	REQUIRED OUTPUT
GET /health	status; modelLoaded; calibrationLoaded; modelVersion
POST /predict-risk	riskLevel; confidence; probabilities; topFactors; modelVersion
POST /predict-batch	Same prediction contract per junction
GET /model-info	Feature schema; target classes; model version; validation summary

```
Frontend
   ↓
Node.js Backend
   ↓
FastAPI
   ↓
Preprocessing + Calibration
   ↓
Selected Model
   ↓
Risk Class + Confidence + Factors
   ↓
Dashboard
```

UNCERTAIN BEHAVIOR

If required features are missing or calibrated confidence is below threshold, return UNCERTAIN according to the agreed policy. The dashboard must distinguish UNCERTAIN from LOW.

17. Updated Team Dependency Matrix

ROLE	WHAT AI/ML NEEDS	TIME
Data/Backend	Junction geometry/master + immutable junction_id	H+1
Data/Backend	Historical accidents with reliable timestamps	H+1
Backend/Traffic	Volume, speed, variance, density, queue, vehicle percentages, peak_hour	H+4
Backend/Safety	Overspeeding, braking, lane changes, wrong-side, pedestrian, near misses	H+6
Backend/Weather	Rainfall, visibility, temperature, weather, wet road, surface, lighting	H+6
Optimization	Coverage and routing/ETA	H+10–H+12
Backend	Prediction contract including confidence/UNCERTAIN	H+13
Frontend	Risk class + confidence + factors + UNCERTAIN UI	H+15
Deployment	ML service URL/environment	H+16

18. Testing & Accuracy Checklist

• Verify feature schema at training and inference.

• Verify no future information enters historical rolling features.

• Verify time-based split for known junctions.

• Verify GroupKFold for unseen junctions.

• Verify 5-fold validation where appropriate.

• Compare Logistic Regression, Random Forest, and XGBoost.

• Tune only inside training/validation.

• Measure HIGH-risk precision and recall.

• Generate confusion matrix.

• Calibrate probabilities.

• Freeze confidence and HIGH decision thresholds.

• Evaluate once on untouched final test period.

• Test UNCERTAIN behavior.

• Test missing/invalid features.

• Test model reload after serialization.

• Test end-to-end backend/dashboard integration.

19. Final Evaluation Table

MODEL	ACCURACY	MACRO F1	HIGH PRECISION	HIGH RECALL	ROC-AUC	STATUS
Logistic Regression	TBD	TBD	TBD	TBD	TBD	Baseline
Random Forest	TBD	TBD	TBD	TBD	TBD	Comparison
XGBoost	TBD	TBD	TBD	TBD	TBD	Primary

DO NOT INVENT RESULTS

All metrics remain TBD until the actual training run. The hackathon report must contain measured results from the locked evaluation protocol.

20. Updated Technical Limitations

1. Synthetic data — Prototype training may use simulated historical accident records.
2. Derived labels — LOW/MEDIUM/HIGH may be engineered when authoritative labels are unavailable.
3. Safety target — ≥90% HIGH precision is a validation objective, not a real-world guarantee.
4. Generalization — GroupKFold tests unseen-junction behavior but cannot replace prospective validation.
5. Data quality — Missing or delayed live traffic/safety/weather feeds can reduce prediction quality.
6. Calibration — Confidence is meaningful only if calibration is verified.
7. Drift — Traffic patterns and road conditions change; production requires monitoring and retraining.
8. Human-in-the-loop — AI remains decision support, not an autonomous high-impact decision maker.

21. Final AI Demonstration

STEP	ACTION	DISPLAY
1	Select junction	LOW/MEDIUM/HIGH + confidence
2	Show normal conditions	Probabilities + top factors
3	Trigger high-severity incident	Feature values update
4	Recalculate	Risk class changes if evidence supports it
5	Show confidence	Calibrated confidence visible
6	Create low-confidence case	UNCERTAIN shown
7	Explain	Top 3–5 real contributors
8	Coverage/optimization	Downstream modules consume locationId + risk
9	Human decision	Operator accepts/rejects

```
Incident
   ↓
AI Features
   ↓
Risk Classification
   ↓
Confidence
   ↓
Explanation
   ↓
Coverage
   ↓
Recommendation
   ↓
Human Decision
```

22. Final Definition of Done

```
Feature Schema Locked
        ↓
Target Definition Locked
        ↓
Leakage-Safe Validation
        ↓
3-Model Comparison
        ↓
```

```
5-Fold / Time-Aware Validation
        ↓
Hyperparameter Tuning
        ↓
HIGH Precision / Recall Analysis
        ↓
Probability Calibration
        ↓
Confidence Threshold
        ↓
LOW / MEDIUM / HIGH / UNCERTAIN
        ↓
Explainability
        ↓
FastAPI
        ↓
Backend
        ↓
Dashboard
        ↓
Coverage / Recommendation
```

FINAL AI/ML OBJECTIVE

Build an explainable, calibrated, leakage-safe junction-risk classifier with a measured HIGH-risk precision target of ≥90% on the agreed evaluation protocol, while explicitly exposing low-confidence predictions as UNCERTAIN.

23. Final Handover Checklist

Data & feature readiness

• Final junction feature schema locked

• Historical windows leakage-safe

• All required traffic/safety/weather features mapped

• Class distribution checked

Model & evaluation

• TimeSeriesSplit completed

• GroupKFold completed for unseen junctions

• 5-fold validation completed where appropriate

• Logistic Regression measured

• Random Forest measured

• XGBoost measured

• Hyperparameters tuned

• HIGH precision/recall measured

• Confusion matrix generated

• Calibration completed

• Threshold frozen

• Final test evaluated once

Deployment & demo

• Model/preprocessor/calibration/configs saved

• FastAPI health works

• Single and batch prediction work

• UNCERTAIN behavior tested

• Top factors verified

• Backend/dashboard integration verified

• Fallbacks tested

• No secrets committed

• Final metrics copied into report

• Demo rehearsed three times

• Model version frozen

24. Conclusion & Action Items

This revision makes accuracy improvement an explicit engineering process rather than a model-name choice. The model now benefits from richer junction-level structural, traffic, safety, environmental, and historical features; leakage-safe validation; three-model comparison; controlled tuning; calibrated probabilities; and a confidence-aware output contract.

Immediate action order: lock the feature schema; confirm data availability; implement leakage-safe rolling features; define labels; train Logistic Regression, Random Forest, and XGBoost; validate with the correct temporal/group strategy; tune; select the HIGH threshold; calibrate probabilities; freeze the model; expose confidence and UNCERTAIN through FastAPI; and integrate with the dashboard.

FINAL TEAM MESSAGE

Do not claim improved accuracy until the new pipeline is actually trained and evaluated. The report now contains the strategy, acceptance criteria, and measurement framework required to demonstrate genuine improvement.